

— A TECHNICAL BRIEFING FOR SENIOR GEOLOGISTS

Every layer of evidence, one ranked list of targets, and the *paper trail to defend it.*

TerraPulse takes an exploration project's data — yours and the world's public record — and returns ranked drill targets, each carrying the evidence that produced it and the full lineage needed to stand behind it.

32 FEATURES ACROSS
7 GROUPS

12 INTEGRATED
DATA SOURCES

5 STAGE PIPELINE,
PROVENANCE
WRITTEN AT EVERY
STAGE

— WHY THIS EXISTS

The problem, as exploration teams actually experience it.

Three frictions every senior geologist on a Ghana licence recognises. None of them are about the geology — they are about the data, the reasoning, and the rework. TerraPulse is built against these three.

PROBLEM 01

01

The data doesn't meet.

Magnetics in one format, soil geochem in another, assay certificates as PDFs, imagery and geology somewhere else entirely. Getting them onto one grid is senior GIS work measured in weeks — not the analysis, just the assembly.

DATA INTEGRATION

01

PROBLEM 02

02

The reasoning doesn't survive.

A target is a judgement made by an experienced geologist looking at several screens. When they leave, or the board asks why, the reasoning is gone — only the conclusion remains. A polygon on a map with no chain of evidence behind it.

REASONING LOSS

02

PROBLEM 03

03

The work doesn't re-run.

New assays land and the integration starts again from scratch. The last analysis was a one-off deliverable, not something you can point at fresh data and press a button on. The work is dead the day it ships.

NO REPRODUCIBILITY

03

— CLAIM 01 / 03

Assembly is automatic.

THE PROBLEM

Weeks of GIS assembly before a single number.

On a typical Ashanti or Sefwi licence, getting magnetics, soil geochem, assay PDFs, imagery and the geological backdrop onto one grid is senior GIS work measured in weeks. It is not the analysis — it is the price of admission to the analysis. And it has to be redone every time a new dataset arrives.

LATENCY: WEEKS · RE-RUN: MANUAL

THE POINT

Upload what you hold. The platform fetches the rest to your boundary.

Sentinel-2

Iron oxide, clay/hydroxyl, vegetation indices

Copernicus DEM

Terrain, slope, drainage, access

NOAA EMAG2

Regional magnetic anomaly, basement context

Macrostrat

Lithology, age, structural units

OneGeology / BRGM

Geological map interpretation backdrop

USGS MRDS

Known deposits, occurrences, claim verification

WDPA

Protected-area exclusion constraint

OpenStreetMap

Road network, settlements, logistics

Your boundary polygon scopes every fetch. What took weeks of GIS assembly is now a boundary and a wait — your survey data and the world's public record on the same grid.

— CLAIM 02 / 03

The ranking explains itself.

THE PROBLEM

A heatmap is a judgement you have to interpret.

Conventional prospectivity output is a colour ramp over a grid. A senior geologist reads it with their own mental model of what drives the score — structural intersection, magnetic low, gold pathfinder — but the map cannot tell them which layer pushed any given cell up. When the board asks why a target is a target, the answer is "it looks red."

OUTPUT: COLOUR RAMP · AUDITABILITY: NONE

THE POINT

Every target carries per-target evidence attribution.

TARGET 03 · RANK 3 / 17 · SCORE 78.4 / 100

PROVENANCE · ON

COORD 6°24'12"N 1°58'40"W · UTM 30N 658420E 713280N

DEPTH Depth to source: ~120 m

EVIDENCE CONTRIBUTION	Δ SCORE
Gold geochemistry (95th pct)	+24.1
Magnetic low / gradient	+18.6
Iron-oxide alteration	+14.3
Clay/hydroxyl alteration	+9.8
Deposit proximity (USGS)	+7.4
Road distance (access)	+4.2

Not a heat map you interpret — a ranked list that tells you what it was reasoning from, with a depth-to-source estimate attached.

THE IMPACT

A geologist can agree or disagree with a specific number, not with a colour

The output is defensible.

THE PROBLEM

The board, the regulator, the QP — they all ask the same question.

NI 43-101, JORC, the Minerals Commission of Ghana — none of them accept a target on faith. “Show your working” is the request, and it arrives at the worst moment: months after the analysis was done, by a different geologist, on data that has since changed. Without lineage attached to the target, the answer is a reconstruction project.

THE POINT

Every layer records its source, processing and parameters.

- 01

Source file + checksum

SHA-256 checksum recorded on arrival, source URL or upload logged per layer.
- 02

Processing steps

Parse → reproject → krigé to 50 m → reduce to pole → depth estimation. Every parameter saved.
- 03

Algorithm version

Deposit model version pinned to the target. Re-runs use the version that produced the original, not the latest.
- 04

QP annotation

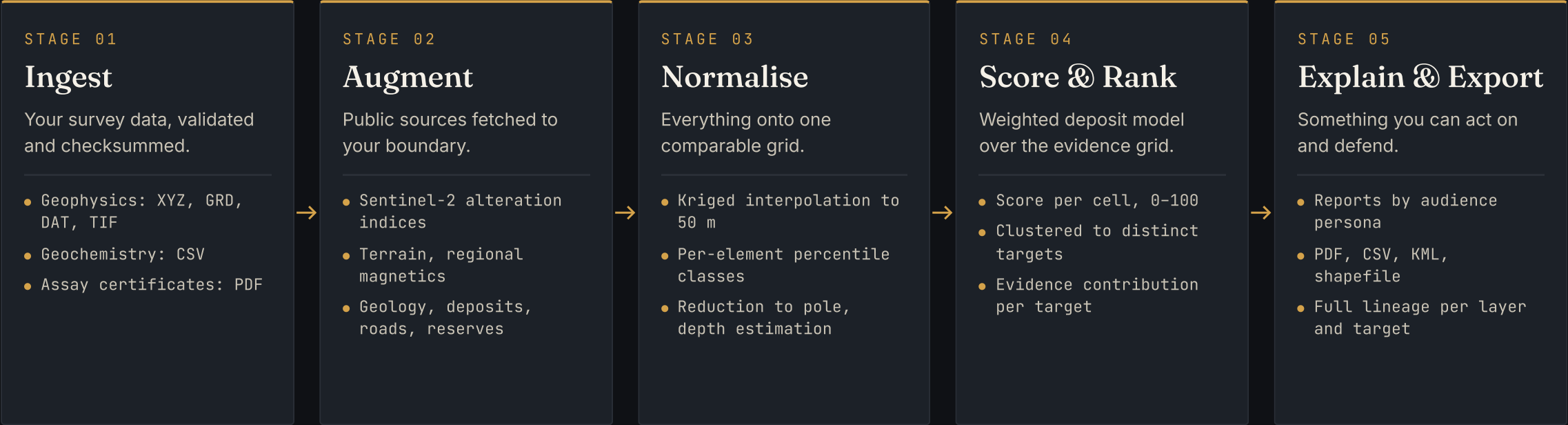
Notes, review status, reviewer name and date attached per target — and carried into every export (PDF, shapefile, KML, CSV).

THE IMPACT

Built for the moment someone asks you to show your working.

HOW IT WORKS, END TO END

One pipeline, five stages. Each stage writes provenance — so the last stage can always answer for the first.



🔗 PROVENANCE WRITTEN AT EVERY STAGE → LAST STAGE CAN ANSWER FOR THE FIRST

The map — where the evidence is read.

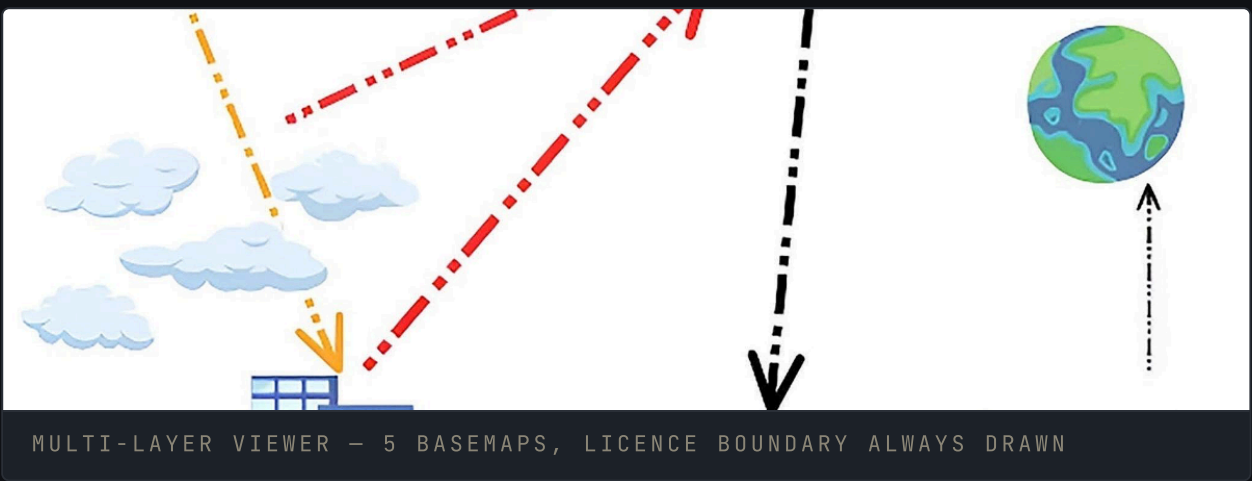
THE PROBLEM

Layers in five different viewers, none of them speaking to each other.

Magnetics in a geophysics package, geochem in a spreadsheet pivot, geology in a PDF, licence boundary in a GIS project file. The geologist keeps four windows open and a mental map of how they align. By the time the question is asked, the alignment was in someone's head.

THE POINT

Every layer on one viewer — visible, faded, interrogated, traced.



- 01 **Multi-layer viewer** — 5 basemaps incl. satellite & 3D terrain; licence boundary always drawn.
- 02 **Layer panel** — visibility & opacity per layer; hover explanation of why each layer matters.
- 03 **Click-to-inspect** — WGS84 + local UTM, raster values, nearest sample & known deposit.
- 04 **Anomaly display** — geochem coloured by percentile class (background → weak → red → strong).

THE IMPACT

The geologist reads the evidence spatially — and the viewer can be traced back to the file that produced it.

Ranked drill targets — the core of the platform.

THE PROBLEM

A target is a judgement — and judgements don’t survive turnover.

On a Ghana licence, a senior geologist picks a drill collar location from experience: this magnetic low, this soil anomaly, this structural intersection. When they move on, the new geologist inherits a polygon, not a reasoning. The next drilling decision starts from a conclusion with no chain behind it.

THE IMPACT

THE POINT

An expert-weighted deposit model over an assembled evidence grid.



PROSPECTIVITY SURFACE — SCORED GRID AS A MAP LAYER, RANKED TARGETS CLUSTERED ON TOP

- 01 **Run analysis** — one action over whatever evidence the project holds; progress reported live.
- 02 **Evidence grid** — per cell: magnetics, alteration, geochem anomaly, lithology, slope, road distance, deposit proximity, protected-area status.
- 03 **Prospectivity scoring** — orogenic-gold model with published weights; 0–100 per cell.
- 04 **Ranked drill targets** — highest cells clustered into distinct targets; rank, score, dual coords, depth-to-source.
- 05 **Provenance panel** — per target: datasets, checksums, processing steps, evidence contributions, model version.

A defensible drill target list — not a polygon someone has to vouch for from memory.

FEATURE GROUP 05 — EXPLORATION COPILOT (V2, 14 TOOLS)

An assistant that can only read your project — and has to cite what it used.

THE PROBLEM

A geologist's questions take days to answer — and the answers don't persist.

"Why is target 03 ranked above target 05?" "What's the As anomaly threshold we used?" "Show me where the alteration signature is strongest." Today each of those is a half-day of re-opening projects, re-running queries, re-plotting. The answer lives in someone's head, then a slide, then nowhere.

THE POINT

14 named tools. Every value resolved server-side. Every claim cited.

Grounded Q&A Answers assembled from your project data, with citations.	Interpretation on request Top-target reasoning, structural setting, data gaps.
Charts from live data Server-resolved values, never written by the model.	Map control from chat Show / hide a layer, the map view changes.
Reads your reports Past conclusions answered from the document, not memory.	Verified claims Deposit names checked against a known-deposits register before display.
Sessions & history Named conversations per project; earlier exchanges as context.	+ 7 MORE 14 named tools total in the Copilot v2 spec.

THE MODEL SELECTS AND INTERPRETS. IT NEVER SUPPLIES VALUES. ANY UNVERIFIED DEPOSIT NAME IS STRIPPED BEFORE DISPLAY.

THE IMPACT

Designed so that it cannot invent a number — the geologist's question gets a defensible answer in seconds.

FEATURE GROUP 06 — REPORTING & EXPORT (V2)

The deliverable — composed from the same data the map uses.

THE PROBLEM

The report and the screen never agree.

A report is written once, exported to PDF, circulated — and is out of date the moment a new assay lands. The map says one thing, the PDF on the regional VP's desk says another. Re-running the analysis doesn't update the report; it just creates another version that nobody reconciles.

THE POINT

5 report types × 5 audience personas — versioned, not overwritten.

REPORT TYPES

Technical Exploration

Executive Summary

Drill Program

Data QA

Comprehensive

AUDIENCE PERSONAS

Exploration Geologist

Qualified Person (NI 43-101 / JORC)

Investor

Project Manager

Environmental Officer

Versioned, not overwritten	Regenerating produces a new version; the old one is preserved.
Section lifecycle	Sections move draft → review → final; human edits are recorded.
Export formats	PDF for circulation; shapefile & KML for GIS / handhelds; CSV for coordinates & scores.
White-label	Consultants put their own logo, company name and contact on every document.

QP & INVESTOR VOICES ADDITIONALLY STRIP UNSUPPORTED CLAIMS

THE IMPACT

The document and the screen can never disagree — they are the same data, written twice.

EVIDENCE & CONSTRAINTS

Where the data comes from — and the four constraints that make the output usable in disclosure.

Twelve sources, all licensed for commercial use, fetched to your boundary.

SOURCE	PROVIDES	ROLE IN SCORING
Your survey data	Magnetics, geochem, assays	Primary anomaly evidence
Sentinel-2	Iron oxide, clay/hydroxyl, vegetation	Alteration signature
Copernicus DEM	Slope, hillshade, drainage	Terrain & access
NOAA EMAG2	Regional magnetic anomaly	Basement & structural context
Macrostrat	Lithology, age, structure	Host-rock favourability
OneGeology / BRGM	Geological map context	Interpretation backdrop
USGS MRDS	Known deposits & occurrences	Proximity weighting
WDPA	Conservation boundaries	Exclusion constraint
OpenStreetMap	Roads, settlements	Access & logistics scoring
Landsat / ASTER	Long-archive SWIR	Change detection, alteration

12 SOURCES TOTAL — rows above + your licence-boundary polygon & project metadata (commercial shape) = 12 integrated inputs.

The argument that matters to a technical buyer.

- 01

The model cannot write a number.

Every figure and statistic in Copilot & reports is resolved from the database server-side. The LLM selects and interprets; it never supplies values.
- 02

Ranking is withheld where the model does not apply.

The deposit model is tuned for orogenic gold. Other commodities get full data, maps and layers — but no ranked targets, rather than numbers from the wrong model.
- 03

Deposit names are verified before they are shown.

Any analogue cited in a generated summary is checked against a known-deposits register, and stripped if it does not match.
- 04

Nothing is a black box.

Weights are published. Contributions are per-target. Processing parameters are recorded. Source files are checksummed.

WHAT IT DOESN'T DO YET — & NEXT STEP

Say these before they're found.

THE PROBLEM

Overselling reads as engineering judgement lost.

The fastest way to lose a senior geologist's trust is to let them discover a limit during the trial. So: the ranking is gold-only — copper, lithium, nickel get every layer, map and geochemical treatment, but no ranked targets. The weights are tuned for orogenic gold. Returning numbers from the wrong model would be worse than returning none.

- GOLD-ONLY RANKING — PER-COMMODITY PROFILES SPECIFIED, NOT YET BUILT
- LICENCE BOUNDARY REQUIRED — NO POLYGON, NO ANALYSIS
- SMALLER GAPS — SOCIAL SIGN-IN, SELF-SERVICE PASSWORD, BRANDING ACCENT — VISIBLE BUT NOT FUNCTIONAL

THE POINT

A Pilot plan engagement — sized for a real evaluation.

SEATS	10	PROJECTS	15
RUNS / MO	150	REPORTS / MO	60
INCLUDES			
Map, upload, analysis, reports + Copilot + team			

Near-Pro capability, sized for a real evaluation rather than a trial. Heavy-AI work draws on usage credits — pay for what you use rather than being throttled. Top-ups via Paystack; visible ledger.

THE IMPACT — NEXT STEP

Bring us one licence polygon on a Birimian ground. We'll return the ranked targets, the evidence attribution per target, and the provenance trail — and a senior geologist on your team decides whether the numbers are defensible.